TEST.0019 Rev. 1.00

Test Procedure – PCB Board – 13300-SE sensor

Revision	Author	Version Date
1.00	Akash Patel	April 6, 2026

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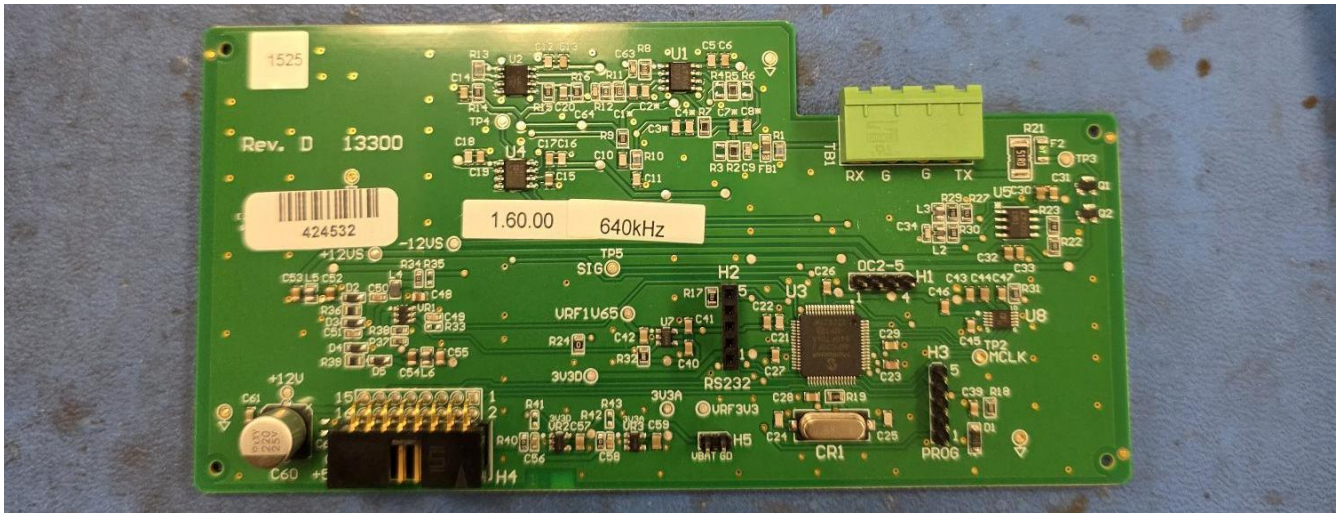
1 Introduction

The purpose of this document is to define the testing process for the **13300-SE sensor board**. This board operates at a frequency of **640kHz**.



The procedure is the same for PCB 13300 operating at a frequency of 1.28 MHz. The differences are in the firmware version, simulator, and labels. Refer to the following table for more information:

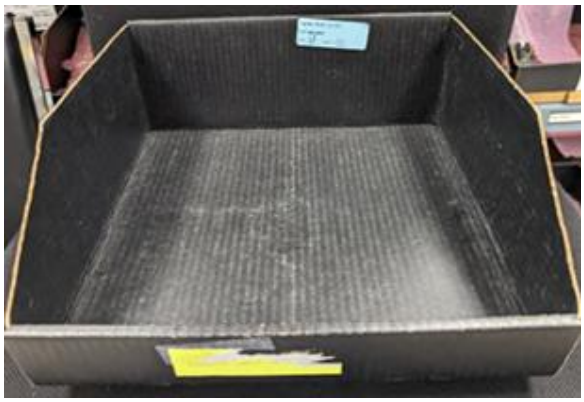
PCB Board	Frequency	Firmware Version	Simulator	Label
13300	640KHz	1.0.1_640KHz and 1.60.0_640KHz	Simulator 640kHz Doppler	1.0.1, 1.60.00, and 640KHz
13300	1.28MHz	1.0.1_1.28MHz and 1.60.0_1.28MHz	Simulator 1.28MHz Doppler	1.0.1, 1.60.00, and 1.28MHz

2 Board Preparation

1. Remove PCB 13300 from the box.

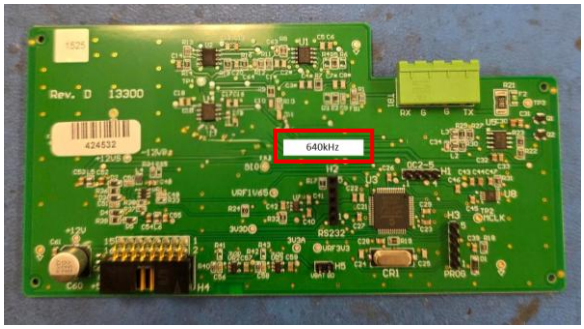
2. Take the bubble wrap off.

3. Place the PCB 13300 in the **Black ESD box**.

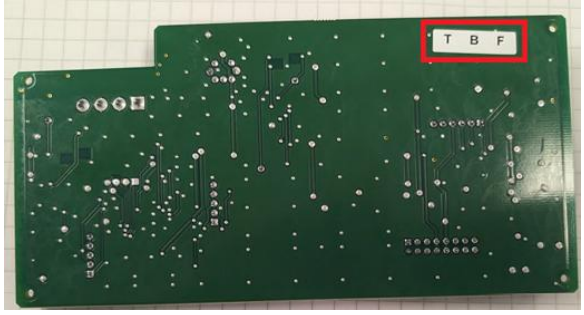


4. Affix Q-number label on the board as shown in the image.

5. Affix **640kHz** label on the board.



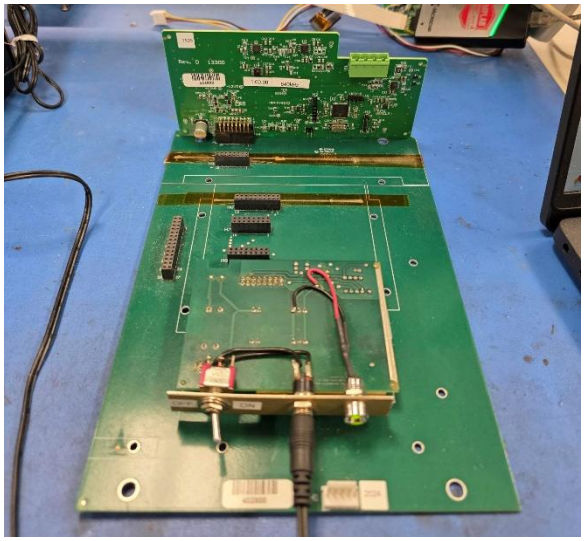
6. Affix test label (**T B F**) on the back of the PCB board. Do not mark anything on the label yet.



3 Install Test Firmware

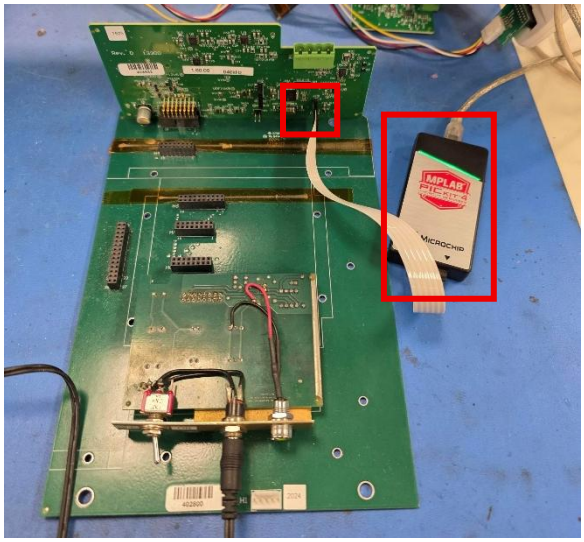
1. Take the programming bench (PCB 13400).

2. Connect PCB 13300 to the programming bench as shown in the image



3. Ensure that the programming bench is switched **ON**.

4. Connect MPLAB cable connector to PCB 13300.

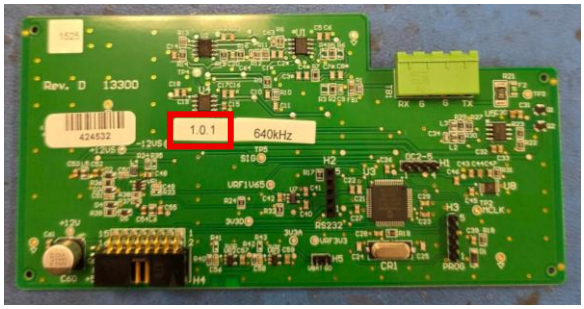


5. Program the firmware version 1.0.1_640KHz. Refer to **SOP001** document for instructions on installing the firmware. The firmware file (as shown in the image) is located at:

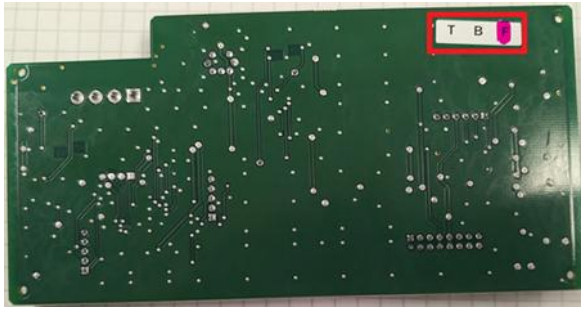
   \\gry_v_s_fil01\Production Data Files\Firmware\GN5\DFM 6.1\13300 BoardTest FW

   Note: Firmware version may change over time.

6. If the program runs successfully, disconnect the board and affix a firmware test label **1.0.1** on the board, as shown in the image.

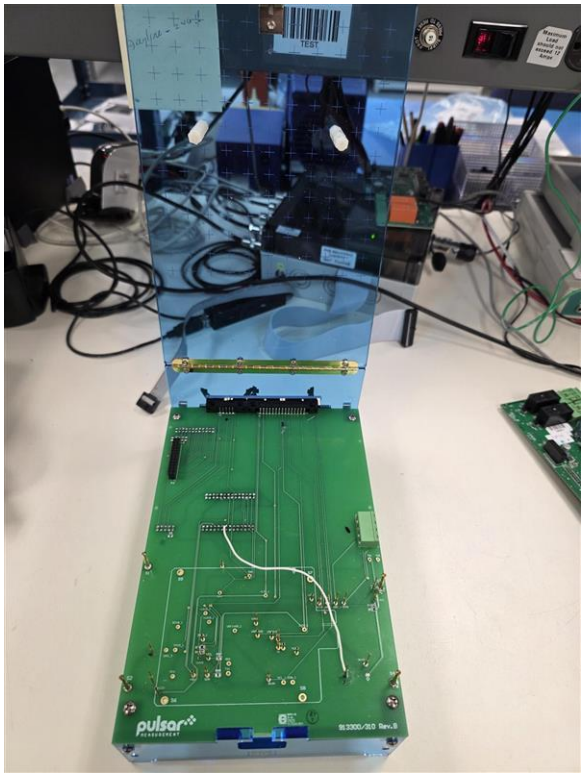


7. If the program fails, disconnect the board and highlight letter **F** on test label. Write **cannot program** on back of the board.

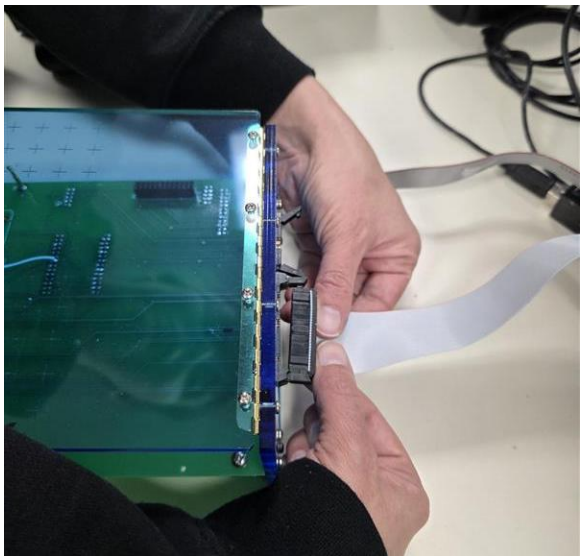


4 Perform Test Procedure

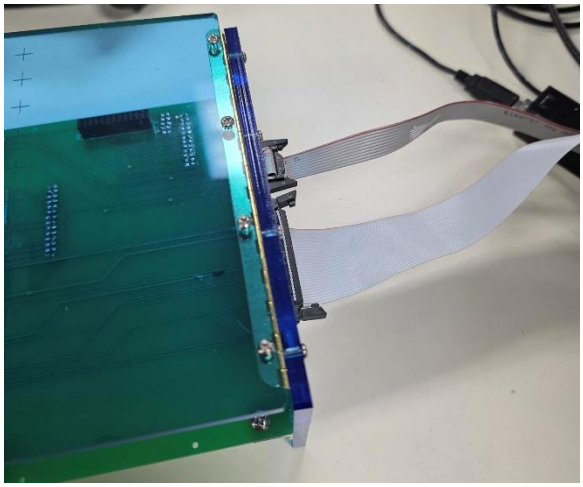
1. Go to the test bench and obtain test board **913300/310** for PCB 13300 testing.



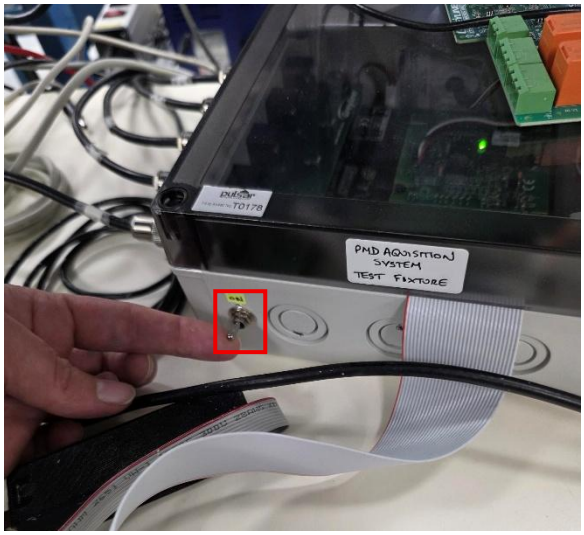
2. Connect the PMD Acquisition System cable to the back of the test board.



3. Connect the UART serial cable.



4. Turn **ON** the PMD Acquisition System.

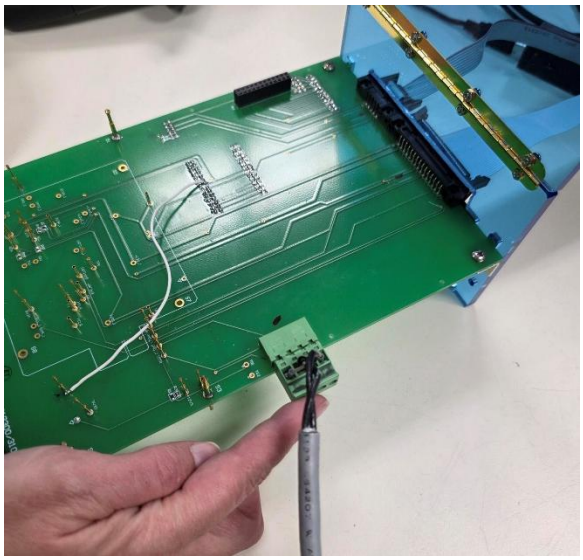


5. For PCB 13300, which operates at 640kHz, use the **640kHz Doppler Simulator**.

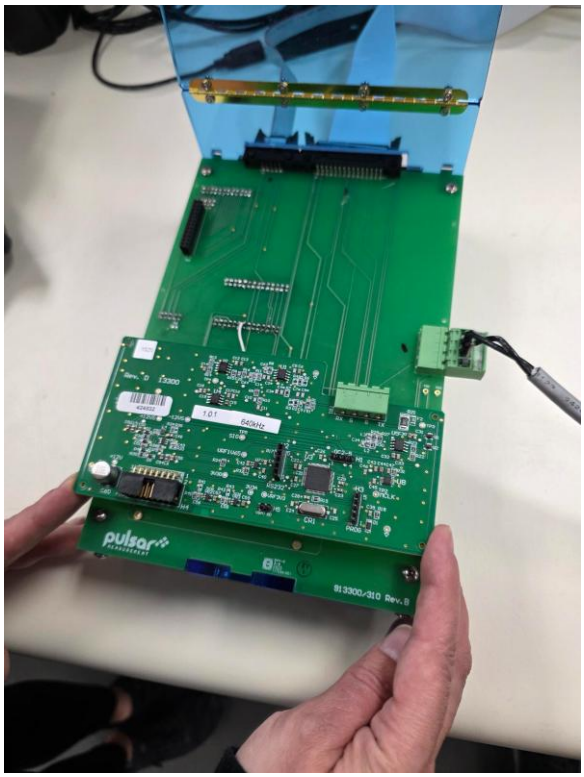
   Note: There are two Doppler Simulators available: 640kHz and 1.28kHz. Use the simulator that matches the PCB board frequency.

6. Get the Doppler simulator's transducer connector.

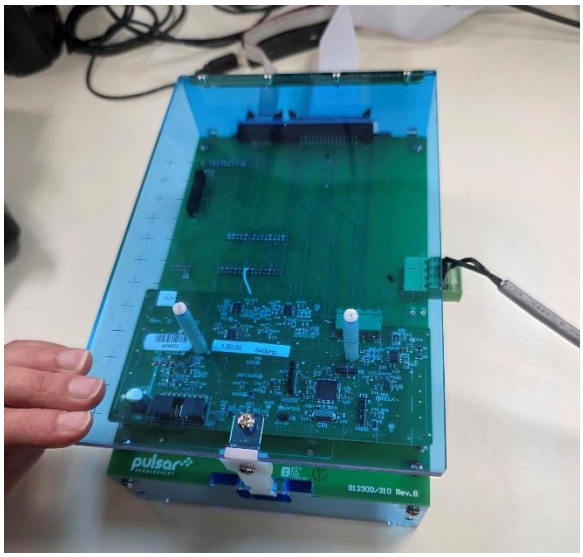
7. Connect it to the PCB board.



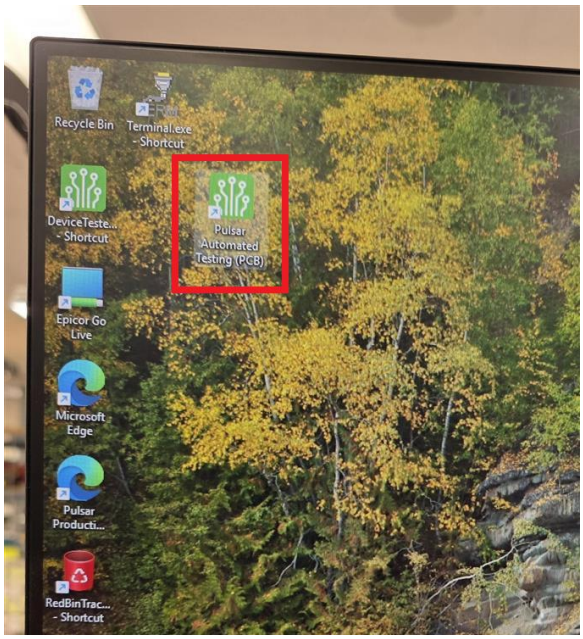
8. Place PCB 13300 on the test board.



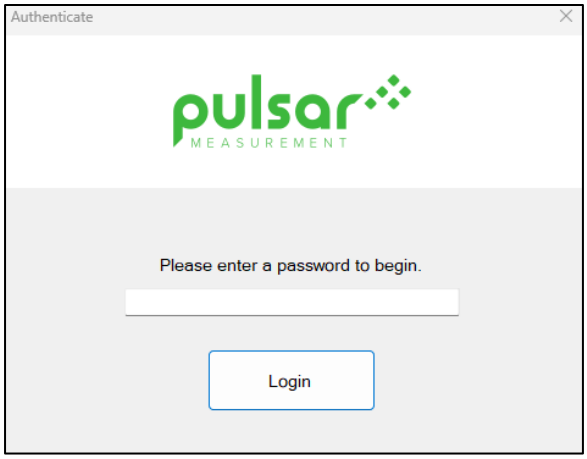
9. Close the lid of the test board.



10. On the Computer, open the **Pulsar Automated Testing (PCB)** application.

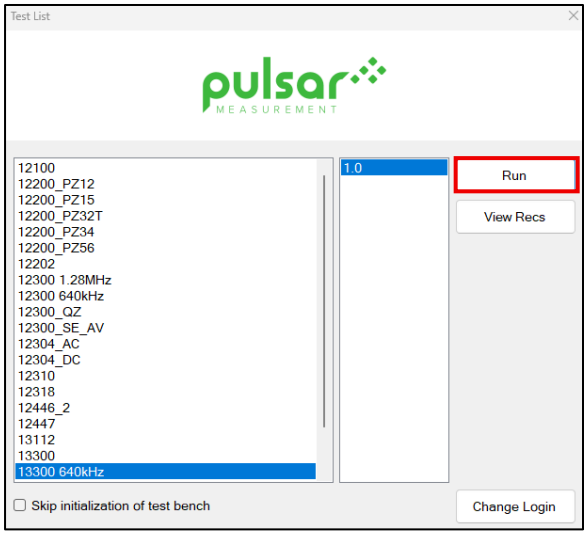


11. The **Authenticate** window opens asking to enter the password.

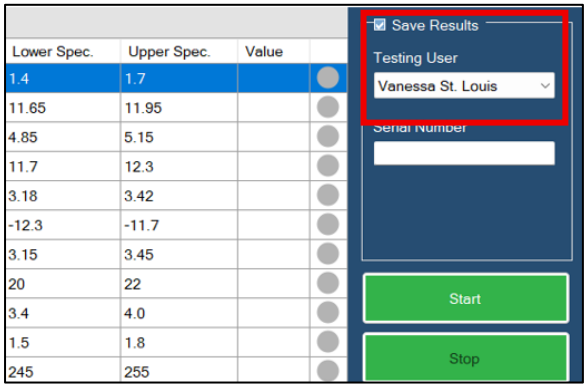


12. Scan the **TEST** label on the shelf with a scan gun. It will authenticate as a password. You can also type **TEST** as a password.

13. In the **Test List** window, select **13300 640kHz > 1.0** and click **Run**.

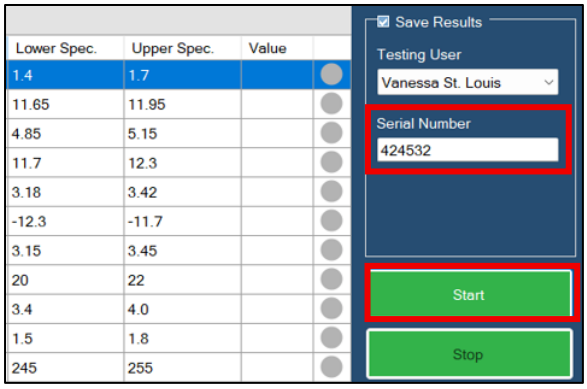


14. Checkmark the **Save Results** checkbox. Select the **Testing User** from the drop-down menu. The last user is selected by default.



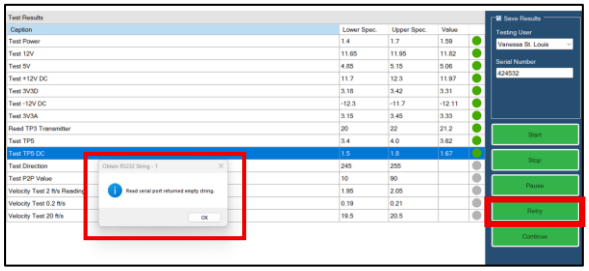
15. To enter the **Serial Number**, scan the label on PCB 13300 with a scan gun.

16. The serial number gets populated in the textbox highlighted. Click the **Start** button to begin the test.

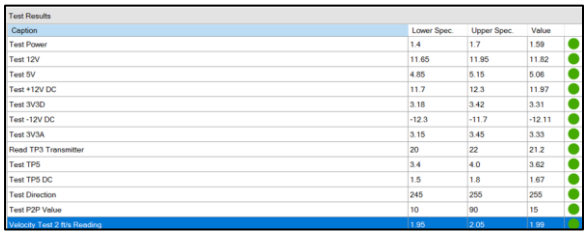


17. If you see an error as "empty string returned", click **OK**. Then click **Retry** to resume the testing.

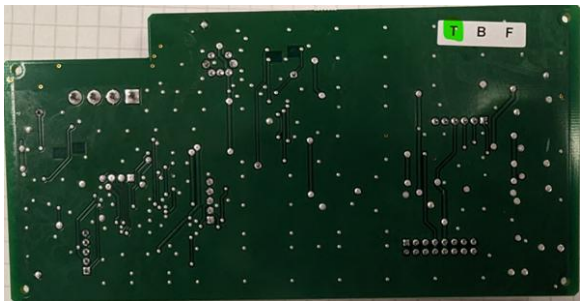
    Note: This error is typically caused by cache memory. You may retry 3–4 times. If the error persists, contact the Engineering Department.



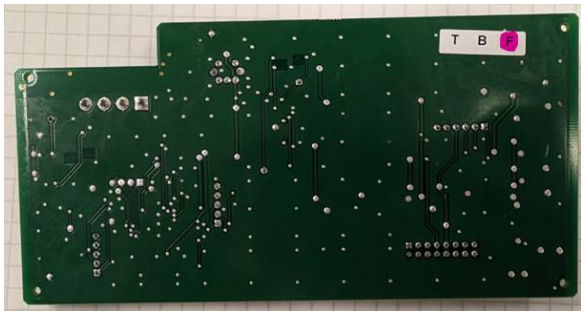
18. The PCB board 13300 passes the testing procedure once all parameters turn green.



19. Once the test is successfully completed, remove the board from the test board and mark the letter **T** on the test label.



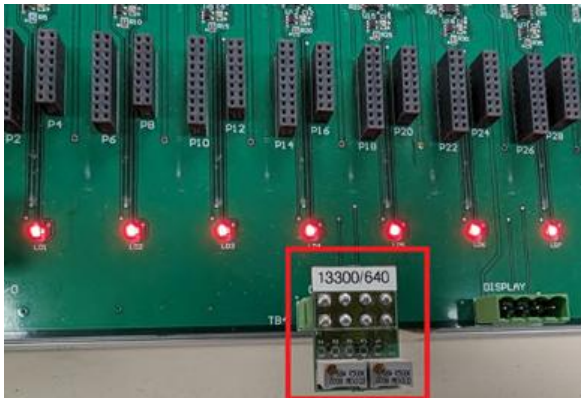
20. If the test fails, remove the board from test board, write reason for failure behind the board and highlight letter **F** on test label.



5 Perform Burn-in Test Procedure

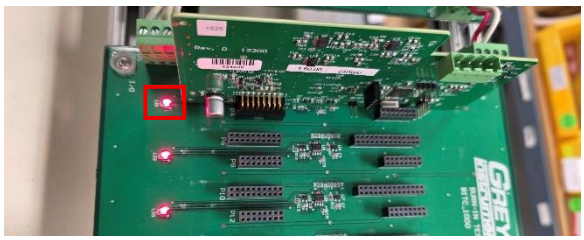
1. Obtain the burn-in reference card and verify that it has the **13300/640** label.

2. Connect the reference card to the burn-in bench, as shown in the image. Ensure it is securely attached.

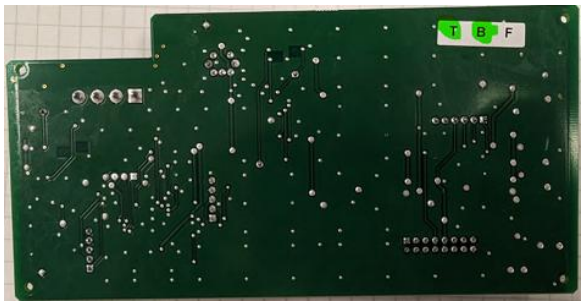


3. Connect PCB 13300 to the burn-in test bench. Ensure the indicator light for the board turns green.

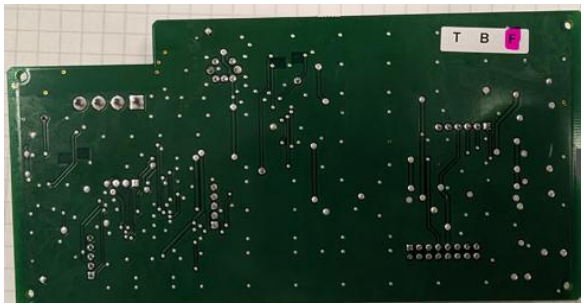
4. If the indicator light turns red, seek assistance from an engineer.



5. Leave the board on the burn-in test bench for 24 hours. After 24 hours, check the indicator light. If the light is green, the board has passed the test.
6. Remove the board from the burn-in test bench and mark the letter **B** on the test label.



7. If the indicator light is red, the board has failed the burn-in test.
8. Remove the board from burn-in test bench and mark the letter **F** on the test label.



9. Place the failed PCB in the **FAILED** board bin.

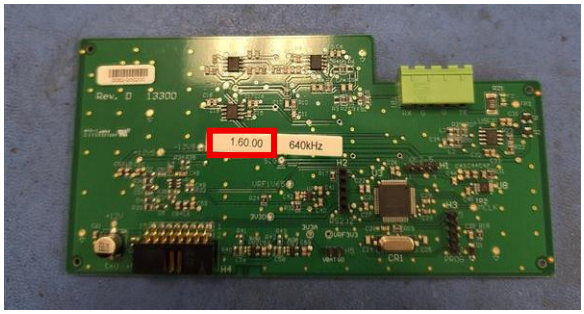
6 Install Production Firmware

1. After performing the burn-in test procedure, reprogram the PCB board to install the production firmware. Follow ***Steps 1-4*** in the Install Test Firmware section.
2. Program the production firmware version 1.60.0_640KHz. Refer to **SOP001** document for instructions on installing the firmware. The firmware file (as shown in the image) is located at:

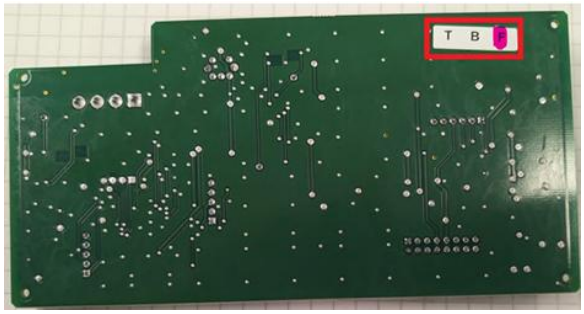
   \\gry_v_s_fil01\Production Data Files\Firmware\GN5\ DFM 6.1\...

   Note: Firmware version may change over time.

3. If the program runs successfully, disconnect the board and affix the production firmware label **1.60.00** on top of the test firmware label, as shown in the image.



4. If the program fails, disconnect the board, mark the letter **F** on the test label, and write **Cannot Program** on the back of the board.



5. Once all tests are completed successfully, place the board in the production bin.

7 Revision Changes

Date	ECN#	Revision	Change
April 6, 2026	N/A	1.00	This is the first release of the document.